

Trigonometry

A Complete Study Guide for Examinations

This guide covers all core trigonometry topics tested in school and university examinations: angles and measurement, trigonometric ratios, identities, graphs, inverse functions, equations, and real-world applications. Work through each section and use the TutorCrew quiz feature to test your understanding.

1. Angles and Measurement

1.1 Degrees and Radians

An angle measures the amount of rotation between two rays sharing a common endpoint (the vertex). Two common units are used:

- Degrees: A full rotation is 360° . A right angle is 90° . A straight angle is 180° .
- Radians: A full rotation is 2π radians. One radian is the angle subtended when the arc length equals the radius.

Conversion formulas:

Degrees to Radians: $\text{radians} = \text{degrees} \times (\pi / 180)$

Radians to Degrees: $\text{degrees} = \text{radians} \times (180 / \pi)$

Common Angle Conversions

Degrees	0°	30°	45°	60°	90°	120°	180°	270°	360°
Radians	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	π	$3\pi/2$	2π

1.2 Arc Length and Sector Area

For a circle of radius r and central angle θ (in radians):

Arc Length: $s = r \times \theta$

Sector Area: $A = (1/2) \times r^2 \times \theta$

Note: Always convert degrees to radians before using these formulas.

2. Trigonometric Ratios

For a right-angled triangle with angle θ , hypotenuse (H), opposite side (O), and adjacent side (A):

$$\sin(\theta) = \text{Opposite} / \text{Hypotenuse} = O / H$$

$$\cos(\theta) = \text{Adjacent} / \text{Hypotenuse} = A / H$$

$$\tan(\theta) = \text{Opposite} / \text{Adjacent} = O / A$$

The reciprocal trigonometric functions are:

$$\csc(\theta) = 1 / \sin(\theta) = H / O \text{ (cosecant)}$$

$$\sec(\theta) = 1 / \cos(\theta) = H / A \text{ (secant)}$$

$$\cot(\theta) = 1 / \tan(\theta) = A / O \text{ (cotangent)}$$

Note: Memory aid: SOH-CAH-TOA (Sine=Opposite/Hypotenuse, Cosine=Adjacent/Hypotenuse, Tangent=Opposite/Adjacent)

2.1 Exact Values of Common Angles

Angle (θ)	$\sin \theta$	$\cos \theta$	$\tan \theta$
0° (0)	0	1	0
30° ($\pi/6$)	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3}$
45° ($\pi/4$)	$1/\sqrt{2}$	$1/\sqrt{2}$	1
60° ($\pi/3$)	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90° ($\pi/2$)	1	0	undefined
180° (π)	0	-1	0
270° ($3\pi/2$)	-1	0	undefined
360° (2π)	0	1	0

2.2 The Unit Circle

The unit circle is a circle of radius 1 centred at the origin. For any angle θ measured from the positive x-axis:

- The point on the unit circle is $(\cos \theta, \sin \theta)$.
- Angles in standard position are measured counterclockwise from the positive x-axis.
- The equation of the unit circle is: $x^2 + y^2 = 1$

Using the unit circle makes it easy to find trig values for any angle, including those greater than 90° or negative angles.

2.3 Signs in the Four Quadrants (CAST Rule)

The sign of each trig function depends on the quadrant the angle lies in:

Quadrant	Angles	sin	cos	tan
I (All positive)	$0^\circ - 90^\circ$	+	+	+
II (Sin positive)	$90^\circ - 180^\circ$	+	-	-
III (Tan positive)	$180^\circ - 270^\circ$	-	-	+
IV (Cos positive)	$270^\circ - 360^\circ$	-	+	-

Note: Memory aid: CAST — going counterclockwise from Quadrant IV: Cos, All, Sin, Tan are positive.

3. Trigonometric Identities

3.1 Pythagorean Identities

These are the most important identities, derived directly from the Pythagorean theorem applied to the unit circle:

$$\sin^2(\theta) + \cos^2(\theta) = 1 \text{ [fundamental]}$$

$$1 + \tan^2(\theta) = \sec^2(\theta) \text{ [divide by } \cos^2\theta]$$

$$1 + \cot^2(\theta) = \csc^2(\theta) \text{ [divide by } \sin^2\theta]$$

3.2 Quotient Identities

$$\tan(\theta) = \sin(\theta) / \cos(\theta)$$

$$\cot(\theta) = \cos(\theta) / \sin(\theta)$$

3.3 Co-function Identities

These show the relationship between complementary angles (angles that add to 90°):

$$\sin(\theta) = \cos(90^\circ - \theta)$$

$$\cos(\theta) = \sin(90^\circ - \theta)$$

$$\tan(\theta) = \cot(90^\circ - \theta)$$

$$\sec(\theta) = \csc(90^\circ - \theta)$$

3.4 Even/Odd (Negative Angle) Identities

Cosine is an even function; sine and tangent are odd functions:

$$\sin(-\theta) = -\sin(\theta) \text{ [odd]}$$

$$\cos(-\theta) = \cos(\theta) \text{ [even]}$$

$$\tan(-\theta) = -\tan(\theta) \text{ [odd]}$$

3.5 Sum and Difference Formulas

These allow us to find the trig value of sums or differences of angles:

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A + B) = (\tan A + \tan B) / (1 - \tan A \tan B)$$

$$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$$

3.6 Double Angle Formulas

Special case of sum formulas where $A = B = \theta$:

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$

$$= 2\cos^2\theta - 1$$

$$= 1 - 2\sin^2\theta$$

$$\tan(2\theta) = 2 \tan \theta / (1 - \tan^2\theta)$$

3.7 Half Angle Formulas

$$\sin(\theta/2) = \pm\sqrt{(1 - \cos \theta) / 2}$$

$$\cos(\theta/2) = \pm\sqrt{(1 + \cos \theta) / 2}$$

$$\tan(\theta/2) = \pm\sqrt{(1 - \cos \theta) / (1 + \cos \theta)}$$

$$= \sin \theta / (1 + \cos \theta)$$

$$= (1 - \cos \theta) / \sin \theta$$

Note: The $\pm$ sign depends on the quadrant of $\theta/2$.

3.8 Product-to-Sum Formulas

$$\sin A \sin B = (1/2)[\cos(A-B) - \cos(A+B)]$$

$$\cos A \cos B = (1/2)[\cos(A-B) + \cos(A+B)]$$

$$\sin A \cos B = (1/2)[\sin(A+B) + \sin(A-B)]$$

4. Graphs of Trigonometric Functions

4.1 Sine Function: $y = \sin(x)$

- Domain: all real numbers | Range: $[-1, 1]$
- Period: 2π (360°)
- Amplitude: 1
- Key points: $(0,0)$, $(\pi/2, 1)$, $(\pi, 0)$, $(3\pi/2, -1)$, $(2\pi, 0)$
- The sine graph is a smooth wave, starting at the origin and going up first.

4.2 Cosine Function: $y = \cos(x)$

- Domain: all real numbers | Range: $[-1, 1]$
- Period: 2π (360°)
- Amplitude: 1
- Key points: $(0,1)$, $(\pi/2, 0)$, $(\pi, -1)$, $(3\pi/2, 0)$, $(2\pi, 1)$
- The cosine graph is the sine graph shifted left by $\pi/2$.

4.3 Tangent Function: $y = \tan(x)$

- Domain: all real numbers except $x = \pi/2 + n\pi$ (where n is an integer)
- Range: all real numbers $(-\infty, +\infty)$
- Period: π (180°)
- Vertical asymptotes at $x = \pm\pi/2, \pm3\pi/2, \dots$
- The graph passes through $(0, 0)$ and increases steeply near asymptotes.

4.4 Transformations: $y = A \sin(Bx + C) + D$

The general form applies to all trig functions. Each constant transforms the graph:

Constant	Effect	Formula
A	Amplitude (vertical stretch/compression)	Amplitude = $ A $
B	Period (horizontal stretch/compression)	Period = $2\pi / B $
C	Phase shift (horizontal translation)	Phase shift = $-C / B$
D	Vertical shift (midline)	Midline: $y = D$

5. Inverse Trigonometric Functions

Inverse trig functions answer the question: "Which angle gives this trig value?" They are written as $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ (or $\arcsin$, $\arccos$, $\arctan$).

Function	Domain	Range	Read as
$y = \arcsin(x)$	$[-1, 1]$	$[-\pi/2, \pi/2]$	"the angle whose sine is x"
$y = \arccos(x)$	$[-1, 1]$	$[0, \pi]$	"the angle whose cosine is x"
$y = \arctan(x)$	$(-\infty, +\infty)$	$(-\pi/2, \pi/2)$	"the angle whose tangent is x"

Key inverse identity examples:

$$\arcsin(\sin \theta) = \theta \text{ for } \theta \text{ in } [-\pi/2, \pi/2]$$

$$\arccos(\cos \theta) = \theta \text{ for } \theta \text{ in } [0, \pi]$$

$$\arctan(\tan \theta) = \theta \text{ for } \theta \text{ in } (-\pi/2, \pi/2)$$

Note: Inverse trig functions always return a single principal value — the restricted range above defines which one.

6. Solving Trigonometric Equations

6.1 General Strategy

- Isolate the trig function on one side.
- Find the principal solution using inverse trig.
- Use the CAST rule or unit circle to find all solutions in $[0^\circ, 360^\circ]$.
- Add the period ($n \times 360^\circ$ or $n \times 2\pi$) for the general solution.

6.2 General Solutions

$$\sin \theta = k \rightarrow \theta = \arcsin(k) + 360n^\circ \text{ or } \theta = 180^\circ - \arcsin(k) + 360n^\circ$$

$$\cos \theta = k \rightarrow \theta = \pm \arccos(k) + 360n^\circ$$

$$\tan \theta = k \rightarrow \theta = \arctan(k) + 180n^\circ$$

Note: n is any integer. The general solution captures infinitely many solutions.

6.3 Example — Solve $\sin \theta = 0.5$ for $0^\circ \leq \theta < 360^\circ$

Step 1: $\arcsin(0.5) = 30^\circ$ (principal value in Quadrant I)

Step 2: $\sin$ is also positive in Quadrant II

Step 3: $\theta = 30^\circ$ or $\theta = 180^\circ - 30^\circ = 150^\circ$

Answer: $\theta = 30^\circ$ or $\theta = 150^\circ$

7. Sine Rule and Cosine Rule

These rules extend trigonometry to any triangle (not just right-angled). For a triangle with sides a , b , c opposite to angles A , B , C respectively:

7.1 The Sine Rule

$$a / \sin A = b / \sin B = c / \sin C$$

Use the Sine Rule when you know:

- Two angles and one side (AAS or ASA)
- Two sides and a non-included angle (SSA — watch for the ambiguous case)

Note: Ambiguous case (SSA): two triangles may satisfy the given conditions. Always check if a second solution exists.

7.2 The Cosine Rule

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Rearranged to find an angle:

$$\cos A = (b^2 + c^2 - a^2) / (2bc)$$

Use the Cosine Rule when you know:

- Two sides and the included angle (SAS)
- All three sides (SSS)

Note: The Cosine Rule reduces to the Pythagorean theorem when $C = 90^\circ$ ($\cos 90^\circ = 0$).

7.3 Area of a Triangle

$$\text{Area} = (1/2) \times a \times b \times \sin C$$

This formula works for any triangle when you know two sides and the included angle. For a right triangle ($C = 90^\circ$), $\sin 90^\circ = 1$, giving the familiar $\text{Area} = (1/2) \times \text{base} \times \text{height}$.

8. Real-World Applications

8.1 Angles of Elevation and Depression

The angle of elevation is the angle measured upward from horizontal to an object above. The angle of depression is measured downward from horizontal to an object below.

$$\tan(\text{angle of elevation}) = \text{height} / \text{horizontal distance}$$

- Example: A flagpole casts a shadow of 15 m. The angle of elevation to the top is 40°. Height = $15 \times \tan(40^\circ) \approx 12.6$ m.

8.2 Navigation and Bearings

Bearings are measured clockwise from North. The sine and cosine rules are used to find distances and directions.

- A ship travels 30 km on a bearing of 060°, then 20 km on a bearing of 140°. The sine/cosine rules find the return distance and bearing.

8.3 Simple Harmonic Motion

Many physical systems (pendulums, springs, sound waves) follow sinusoidal patterns:

$$x(t) = A \sin(\omega t + \phi)$$

- A = amplitude (maximum displacement)
- ω = angular frequency ($\omega = 2\pi / T$ where T is the period)
- ϕ = phase angle (initial displacement)
- t = time

8.4 Waves and Oscillations

The sine and cosine functions describe all wave-like phenomena including light, sound, AC electricity, and ocean waves.

$$y(x, t) = A \sin(kx - \omega t)$$

- k = wave number = $2\pi / \lambda$ (λ is wavelength)
- ω = angular frequency = $2\pi / T$
- Speed of wave: $v = \omega / k = \lambda / T$

9. Quick Reference Summary

9.1 Must-Know Identities Checklist

- ✓ $\sin^2\theta + \cos^2\theta = 1$
- ✓ $\tan \theta = \sin \theta / \cos \theta$
- ✓ $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
- ✓ $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
- ✓ $\sin(2\theta) = 2 \sin \theta \cos \theta$
- ✓ $\cos(2\theta) = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \cos^2\theta - \sin^2\theta$
- ✓ Sine Rule: $a/\sin A = b/\sin B = c/\sin C$
- ✓ Cosine Rule: $a^2 = b^2 + c^2 - 2bc \cos A$
- ✓ Area = $(1/2)ab \sin C$
- ✓ Arc length $s = r\theta$ (θ in radians)
- ✓ Sector area $A = (1/2)r^2\theta$ (θ in radians)

9.2 Common Mistakes to Avoid

- Forgetting to convert degrees to radians when using arc length / sector area formulas.
- Using SOH-CAH-TOA on a non-right-angled triangle — use Sine/Cosine Rule instead.
- Missing the second solution when solving trig equations (CAST rule).
- Ignoring the ambiguous case in SSA triangles.
- Confusing $\sin^2\theta$ with $\sin(\theta^2)$ — they are completely different expressions.
- Forgetting that $\tan \theta$ is undefined at $\theta = 90^\circ, 270^\circ$, etc.
- Not restricting the range when using inverse trig functions.

End of Trigonometry Study Guide. Upload this PDF to TutorCrew and generate a quiz to test your understanding of all nine sections.